

1. Children of three ages are asked to indicate their preference for three photographs of adults.

Do the data suggest that there is a significant relationship between age and photograph preference? What is wrong with this study? (10 Marks)

Photograph:

Age of child A B C

5-6 years: 18 22 20

7-8 years: 2 28 40

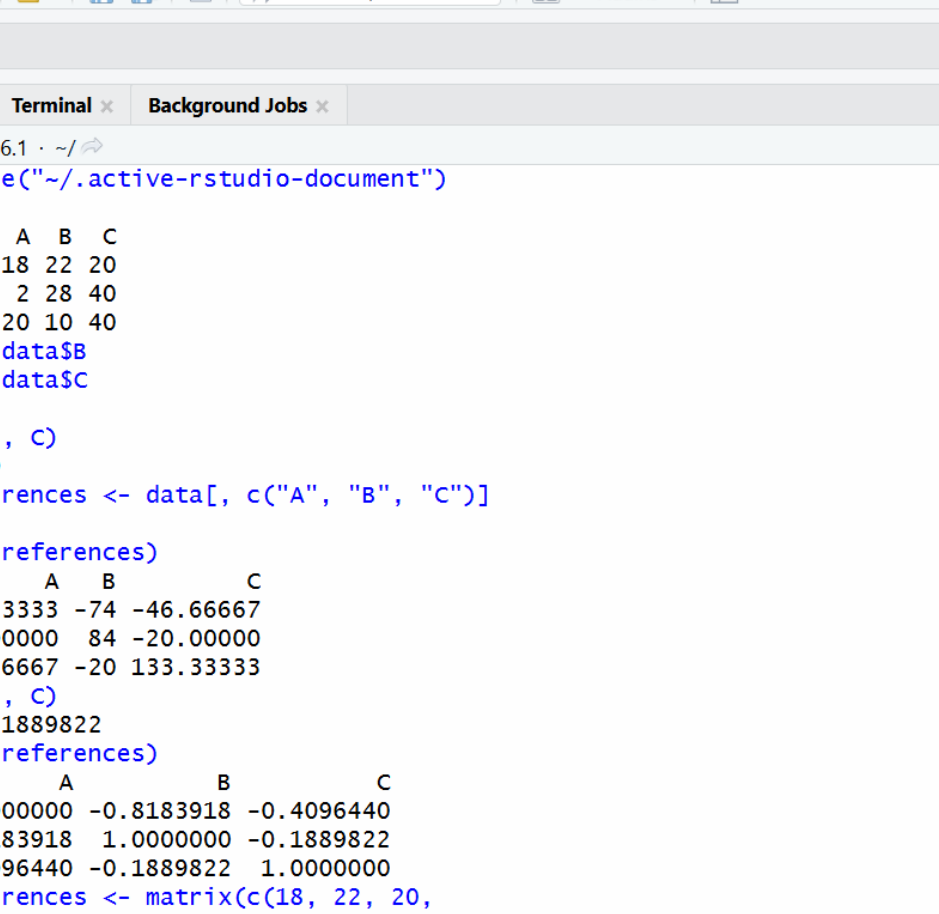
9-10 years: 20 10 40

Use `cov()` to calculate the sample covariance between B and C.

Use another call to `cov()` to calculate the sample covariance matrix for the preferences.

Use `cor()` to calculate the sample correlation between B and C.

Use another call to `cor()` to calculate the sample correlation matrix for the preferences.



The screenshot shows the RStudio interface with the following components:

- Top Bar:** RStudio logo and menu items: File, Edit, Code, View, Plots, Session, Build, Debug, Profile, Tools, Help.
- Toolbar:** Icons for adding files, saving, printing, and a search bar labeled "Go to file/function".
- Source Panel:** A tab labeled "Source" with a file icon.
- Console Panel:** A tab labeled "Console" with a terminal icon. It shows the execution of R code and its output.
- Terminal Panel:** A tab labeled "Terminal" with a terminal icon, currently empty.
- Background Jobs Panel:** A tab labeled "Background Jobs" with a terminal icon, currently empty.

The console output shows the following R code and its results:

```
> source("~/active-rstudio-document")
> data
  Age  A  B  C
1  5-6 18 22 20
2  7-8  2 28 40
3 9-10 20 10 40
> B <- data$B
> C <- data$C
>
> cov(B, C)
[1] -20
> preferences <- data[, c("A", "B", "C")]
>
> cov(preferences)
      A      B      C
A  97.33333 -74 -46.66667
B -74.00000  84 -20.00000
C -46.66667 -20 133.33333
> cor(B, C)
[1] -0.1889822
> cor(preferences)
      A      B      C
A  1.0000000 -0.8183918 -0.4096440
B -0.8183918  1.0000000 -0.1889822
C -0.4096440 -0.1889822  1.0000000
> preferences <- matrix(c(18, 22, 20,
+                          2, 28, 40,
+                          20, 10, 40),
+                        nrow = 3,
+                        byrow = TRUE)
>
```

P5 Player E 18

P6 Player F 20

P7 Player G 22

P8 Player H 13

P9 Player I 11

P10 Player J 35

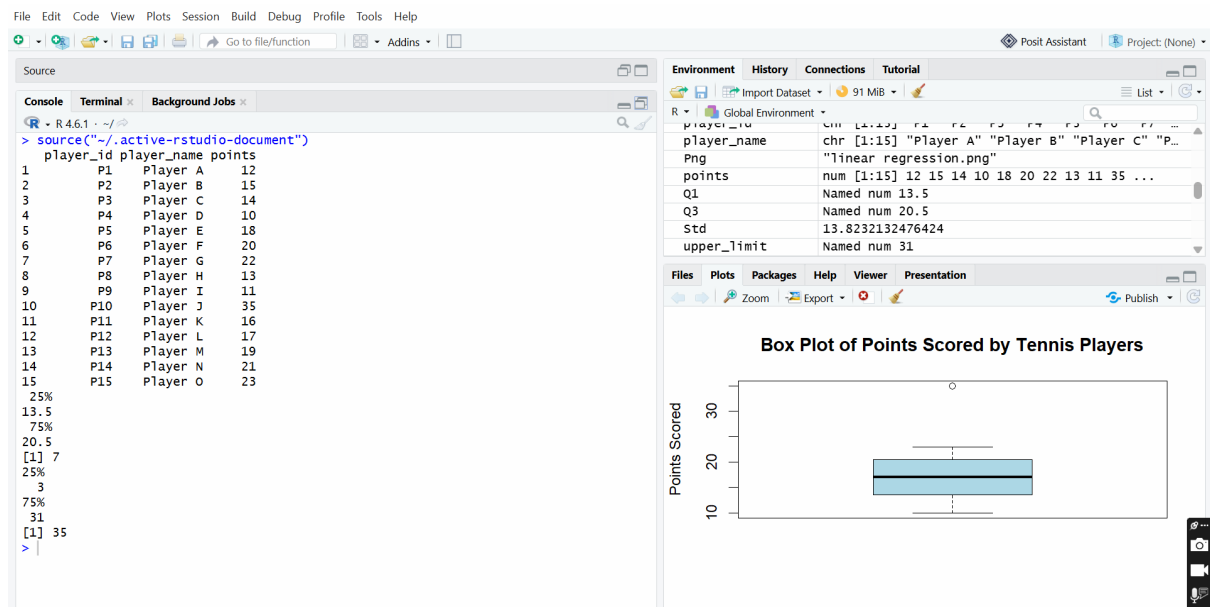
P11 Player K 16

P12 Player L 17

P13 Player M 29

P14 Player N 21

P15 Player O 23



3. A bank wants to decide whether to approve a loan based on customer attributes like:

Age

Income

Employment Status

Credit Score

Using historical data, build a Decision Tree classifier to predict Loan Approval (Yes/No).

(10 Marks)

Consider the Data Set

young,high,employed,good,yes

young,high,self-employed,average,yes

middle,high,employed,good,yes

old,medium,employed,good,yes

old,low,unemployed,poor,no

old,low,self-employed,average,no

middle,low,unemployed,poor,no

young,medium,employed,average,yes

young,low,unemployed,poor,no

old,medium,self-employed,good,yes

middle,medium,employed,average,yes

middle,high,self-employed,good,yes

old,medium,unemployed,poor,no

young,medium,self-employed,average,yes

middle,low,employed,poor,no

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier Choose J48 -C 0.25 -M 2

Test options

- ☒ Use training set
- ☐ Supplied test set Set...
- ☐ Cross-validation Folds 10
- ☐ Percentage split % 66

More options...

(Nom) LoanApproval

Start Stop

Result list (right-click for options)

06:24:48 - trees.J48

Classifier output

```
=== Evaluation on training set ===
Time taken to test model on training data: 0 seconds

=== Summary ===
Correctly Classified Instances      14      93.3333 %
Incorrectly Classified Instances    1      6.6667 %
Kappa statistic                    0.8571
Mean absolute error                 0.1067
Root mean squared error             0.2309
Relative absolute error              22.1138 %
Root relative squared error         47.1269 %
Total Number of Instances          15

=== Detailed Accuracy By Class ===
               TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
               1.000    0.167    0.900     1.000    0.947     0.866    0.963    0.956    yes
               0.833    0.000    1.000     0.833    0.909     0.866    0.963    0.933    no
Weighted Avg.   0.933    0.100    0.940     0.933    0.932     0.866    0.963    0.947

=== Confusion Matrix ===
a b   <-- classified as
9 0 | a = yes
1 5 | b = no
```

Status OK Log

Weka Explorer

reprocess Classify Cluster Associate Select attributes Visualize

Classifier Choose J48 -C 0.25 -M 2

Test options

- ☒ Use training set
- ☐ Supplied test set Set...
- ☐ Cross-validation Folds 10
- ☐ Percentage split % 66

More options...

(Nom) LoanApproval

Start Stop

Result list (right-click for options)

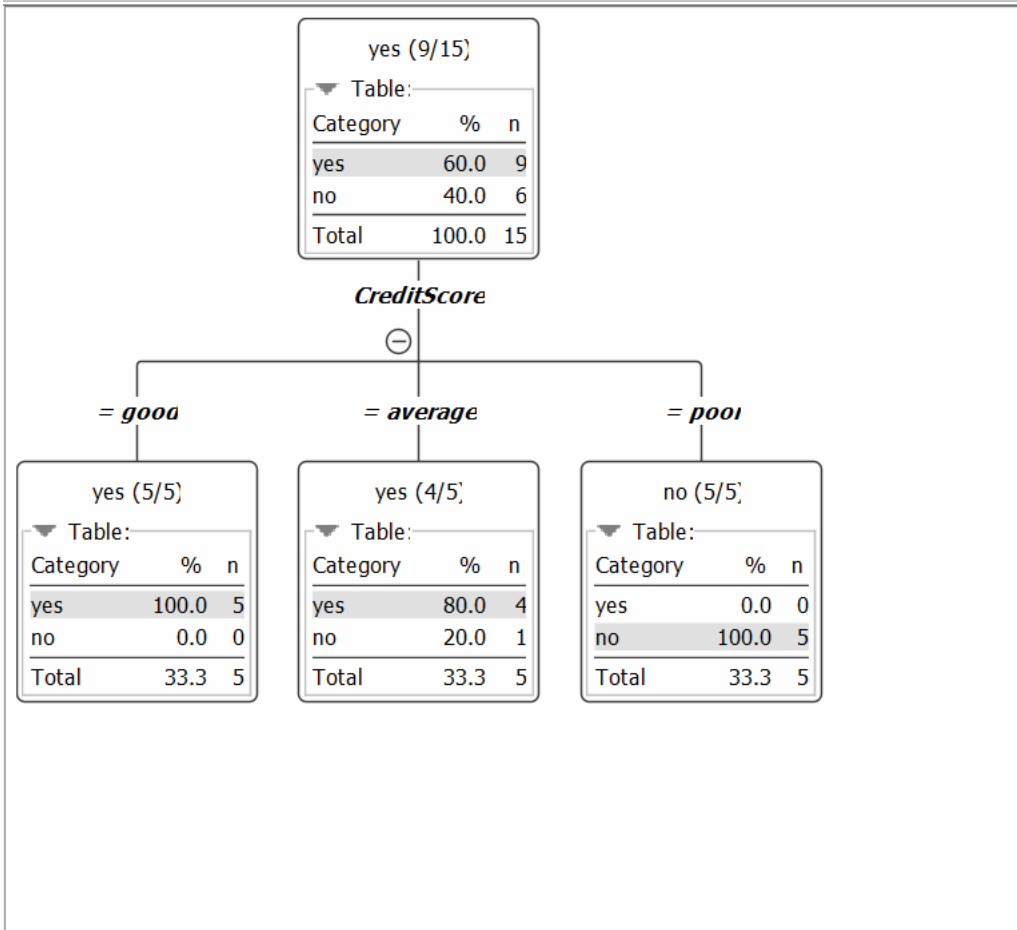
06:24:48 - trees.J48

Weka Classifier Tree Visualizer: 06:24:48 - trees.J48 (loan_approval)

Tree View

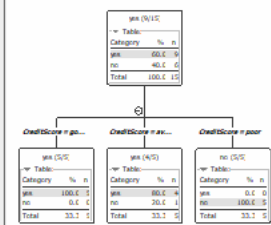
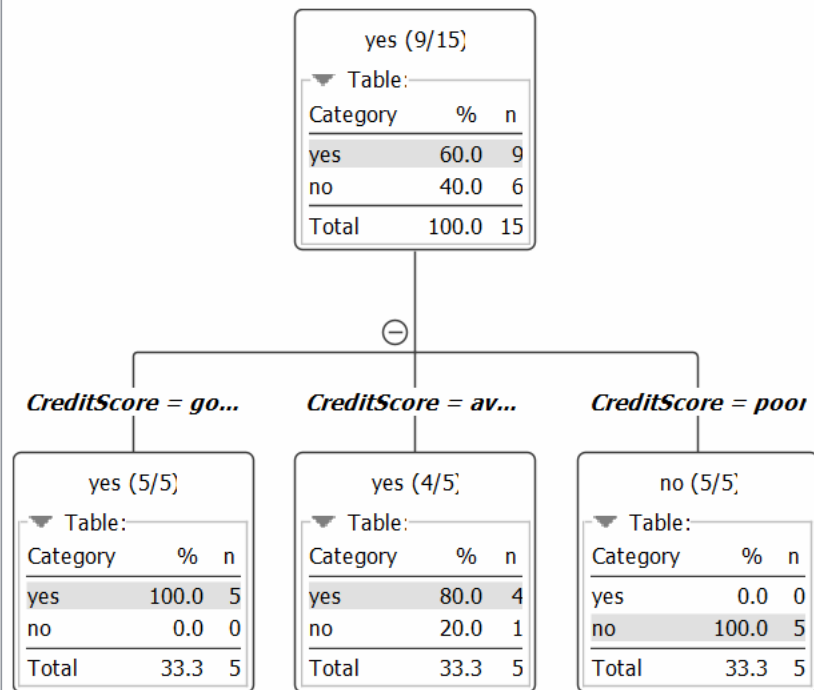
```
graph TD
    CS([CreditScore]) -- "= good" --> Y1[yes (5.0)]
    CS -- "= average" --> Y2[yes (5.0/1.0)]
    CS -- "= poor" --> N1[no (5.0)]
```

Status OK Log



Decision Tree View - 3:3 - Decision Tree Predictor

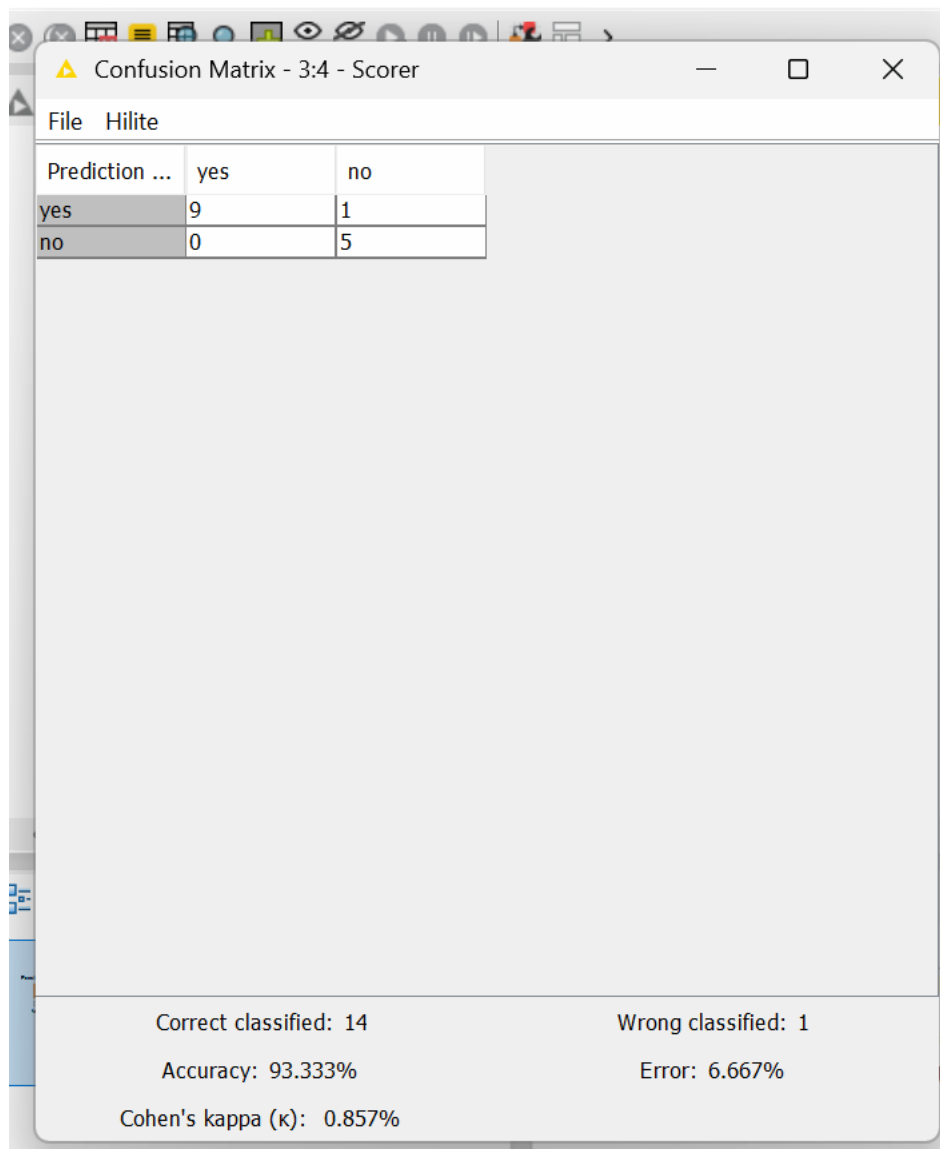
File HiLite Tree



Zoom:

100.0%

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4. Two Maths teachers are comparing how their Year 9 classes performed in the end of year exams. Their results are as follows: (5 Marks)

Class A: 76, 35, 47, 64, 95, 66, 89, 36, 84, 76, 35, 47, 64, 95, 66, 89, 36, 84

Class B: 51, 56, 84, 60, 59, 70, 63, 66, 50, 51, 56, 84, 60, 59, 70, 63, 66, 50

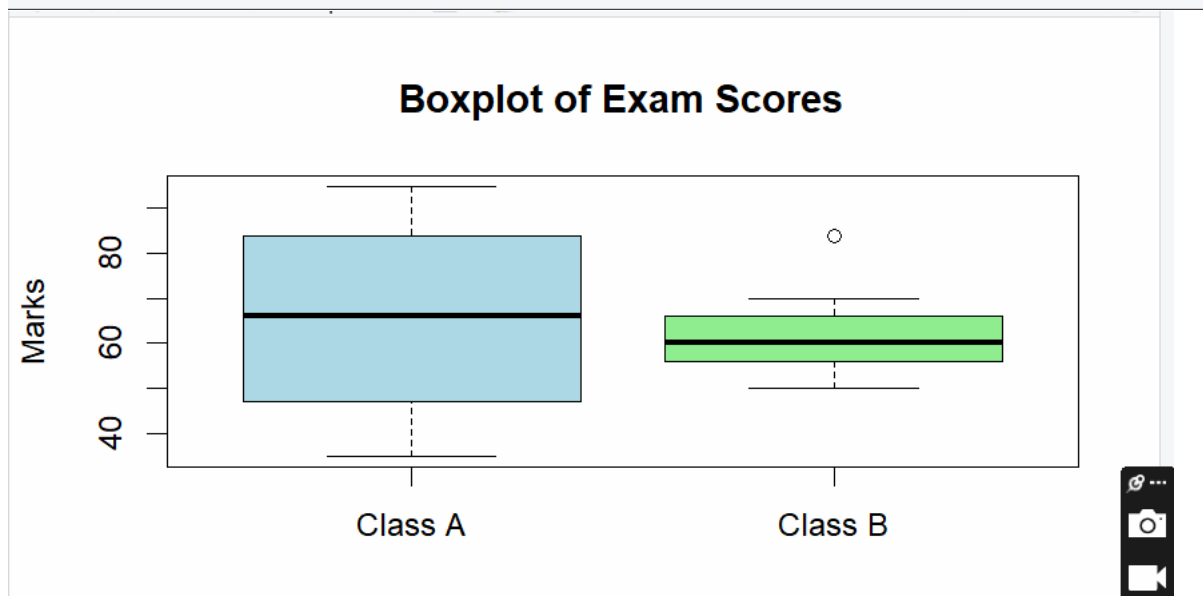
(i) Find which class had scored higher mean, median and range.

(ii) Plot above in boxplot and give the inferences


```

Console Terminal Background Jobs
R 4.6.1 ~ /
> source("~/active-rstudio-document")
Class A Mean = 65.77778
Class B Mean = 62.11111
Class A Median = 66
Class B Median = 60
Class A Range = 60
Class B Range = 34
>

```



5. Suppose that a hospital tested the age and body fat data for 18 randomly selected adults with the following results:

age	23	23	27	27	39	41	47	49	50
%fat	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2
age	52	54	54	56	57	58	58	60	61
%fat	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

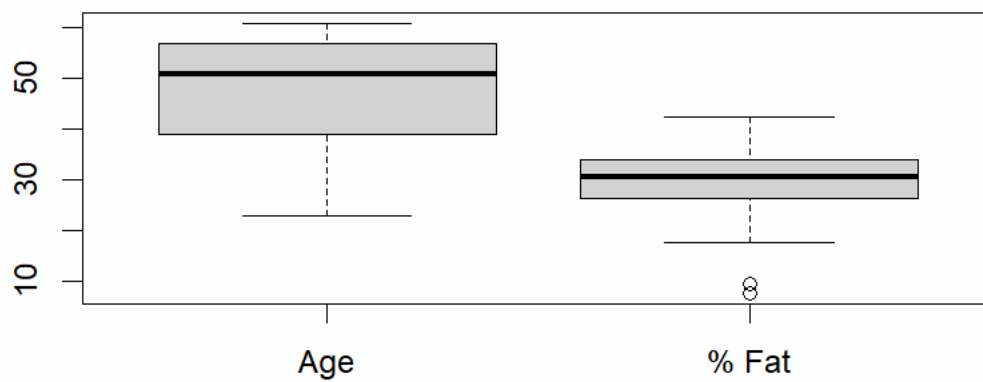
Calculate the mean, median, and standard deviation of age and %fat.

(b) Draw the boxplots for age and %fat.

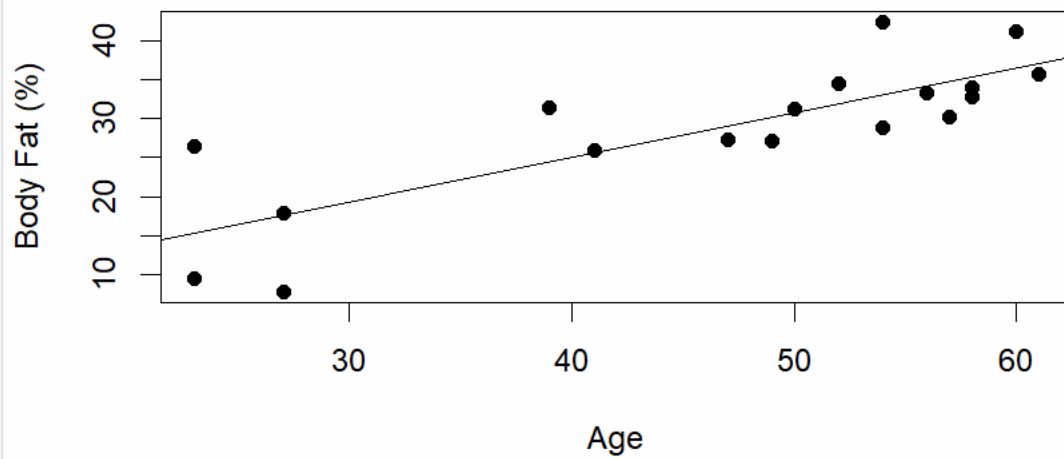
(c) Draw a scatter plot and a q-q plot based on these two variables.

```
R 4.0.1 ~ /  
> source("~/active-rstudio-document")  
> mean(age)  
[1] 46.44444  
> mean(fat)  
[1] 28.78333  
> median(age)  
[1] 51  
> median(fat)  
[1] 30.7  
> sd(age)  
[1] 13.21862  
> sd(fat)  
[1] 9.254395  
> |
```

Boxplots of Age and Body Fat



Scatter Plot: Age vs Body Fat



Q-Q Plot of Age

